Assignment 10

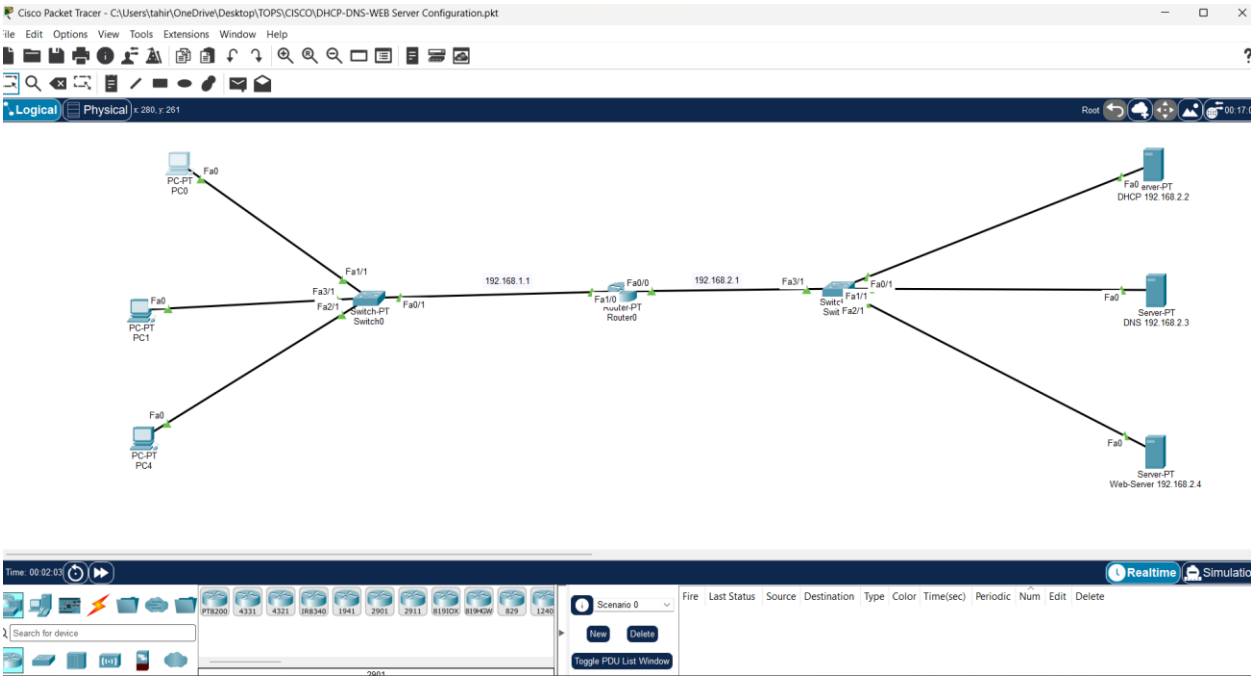
DHCP Configuration

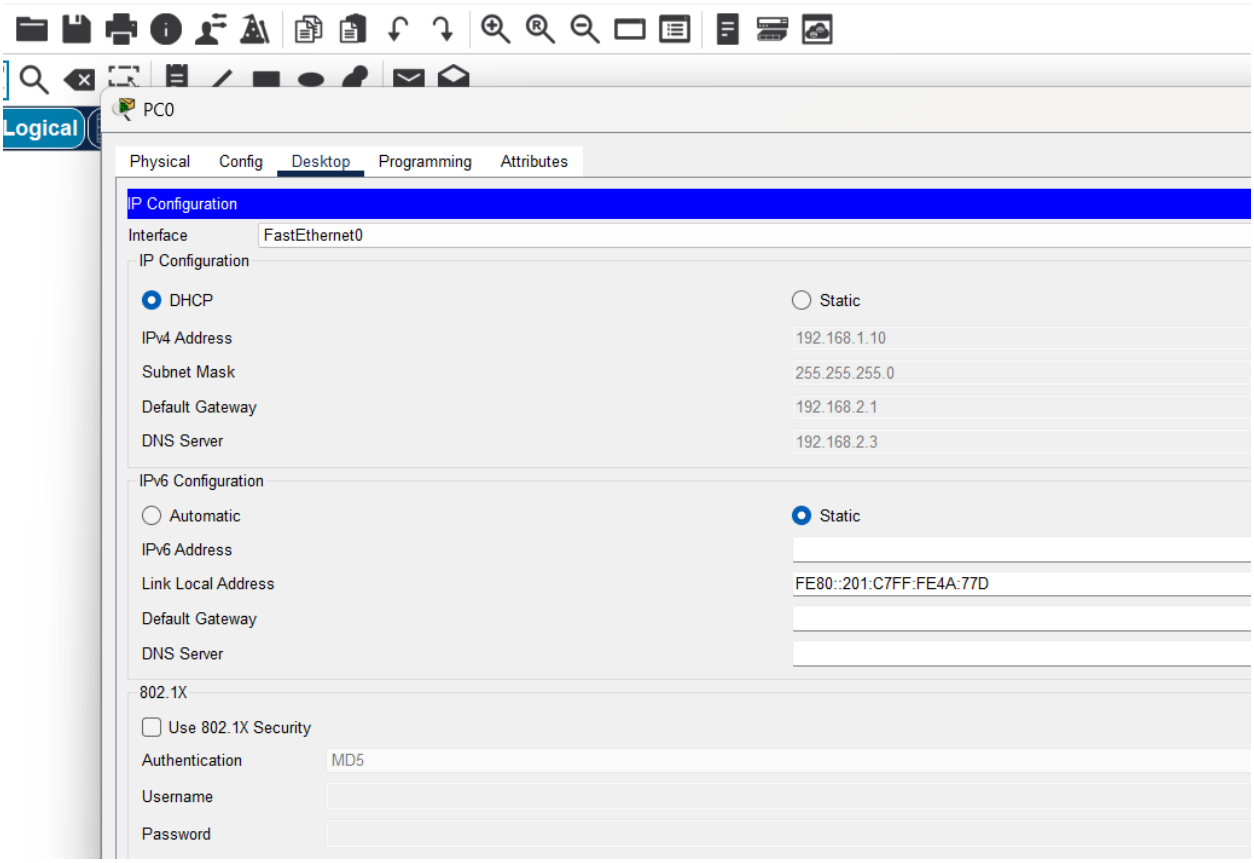
1. **In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.**

Result: In Cisco Packet Tracer, I configured one DHCP server with a starting IP address of 192.168.1.10 and connected two client PCs. Both PCs received different IP addresses automatically from the DHCP pool**.**

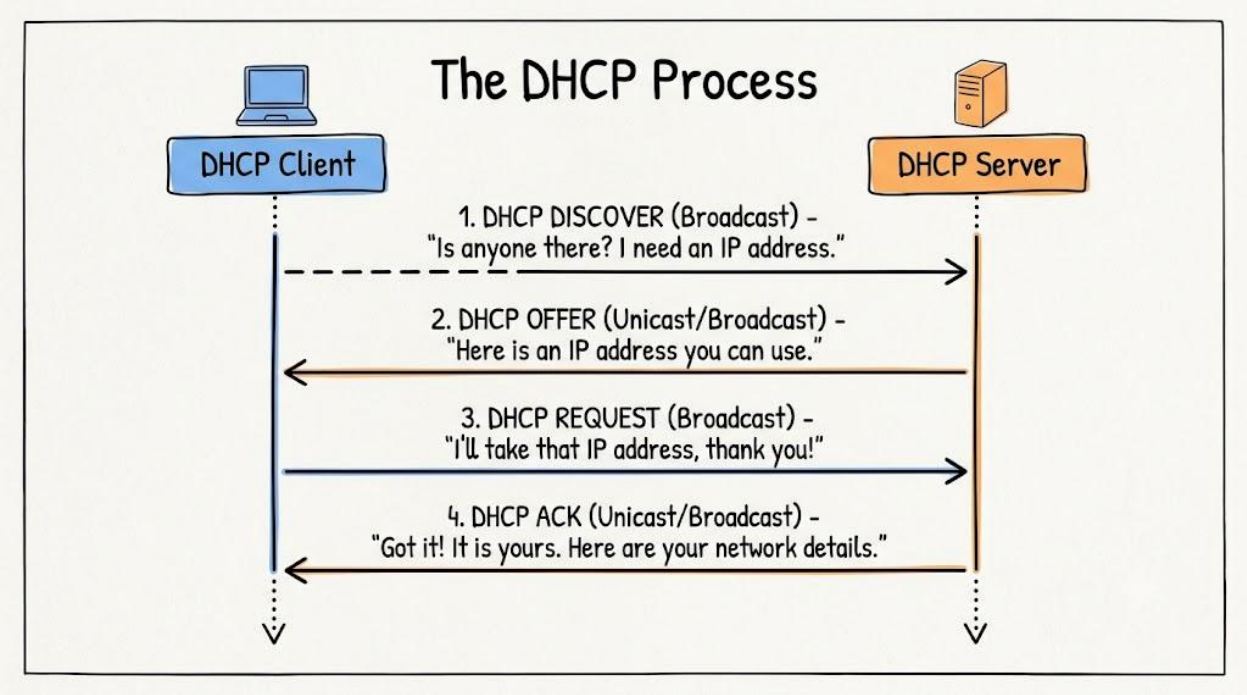
- **PC0: 192.168.1.10**
- **PC1: 192.168.1.11**

❖ **Conclusion:** DHCP successfully assigned unique IP addresses to both PCs automatically, without requiring manual IP configuration.


Cisco Packet Tracer - C:\Users\tahir\OneDrive\Desktop\TOPS\CISCO\DHCP-DNS-WEB Server Configuration.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
PC-PT PC0
PC-PT PC1
PC-PT PC4
Switch-PT Switch0
Router-PT Router0
Switch Swit
Server-PT DHCP 192.168.2.2
Server-PT DNS 192.168.2.3
Server-PT Web-Server 192.168.2.4
192.168.1.1
192.168.2.1
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete


Logical
PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
192.168.2.3
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:C7FF:FE4A:77D
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

2. **Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage. Hint: Use arrows to show the direction of each message between client and server.**



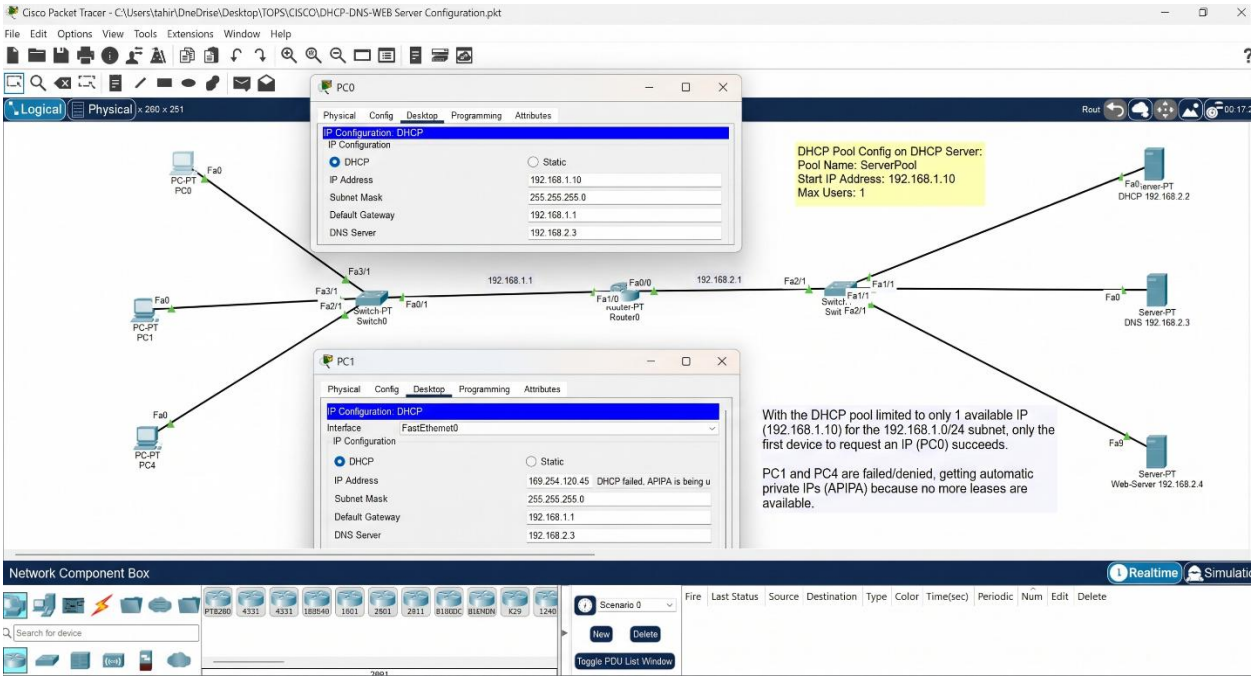
1. **Discover (Client ➡️ Server):** The client broadcasts a message asking if there is an IP address available for use.

2. **Offer (Server ➡️ Client):** A DHCP server responds back with an available IP address option.

3. **Request (Client ➡️ Server):** The client broadcast-requests to accept that offered IP address.

4. **Acknowledge (Server ➡️ Client):** The server confirms the reservation and sends the required network details (subnet mask, DNS, gateway) to finish the setup.

3. **Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.**



4. **Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.**

